

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) Semester IV Examination, 2025-26 (NEP)

Subject: Mathematics

Paper No. MATMJ-42 : Vector and Tensor Analysis

Time: 2½ Hours

Full Marks: 70

REF NO. Conf/Even/2025-26/65

*Instructions: Answer **five** questions including Question 1 which is compulsory. Make proper depiction of lines, curves, surfaces and volumes wherever necessary. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory)

- (a) Make sketch of a simple curve, simple closed curve, simply connected domain and multiply connected domain. [2]
- (b) Find $\text{grad} \left(\frac{1}{r} \right)$, where $\vec{r}$ is the position vector and $r = |\vec{r}|$. [2]
- (c) In which direction from the point $(2, 1, -1)$ is the directional derivative of $\phi = x^2yz^3$ maximum and find its magnitude. [2]
- (d) Write down the expression for the Christoffel symbols of first and second kind $[pq, r]$ and $\left\{ \begin{smallmatrix} s \\ pq \end{smallmatrix} \right\}$ respectively, where p, q, r, s denotes indices. [2]
- (e) Prove that the contraction of the tensor A_q^p is a scalar or invariant. [2]
- (f) Find the area of the region bounded by $y^3 = x$, the y -axis and the lines $y = 3$ and $y = 6$. [3]
- (g) Find $\text{curl} \left(\frac{a\vec{r}}{r^3} \right)$ where a is a scalar and $\vec{r}$ is the position vector and $r = |\vec{r}|$. Is this field irrotational? [3]
- (h) Show that $\iiint_V \text{grad} \phi dV = \iint_S \phi \hat{n} dS$ where V is the volume enclosed by the surface S , ϕ is a scalar function and $\hat{n}$ is the unit outward normal on S . [3]
- (i) If A_i and B^i are the components of a covariant and contravariant vector respectively, then show that $A_i B^i$ is an invariant. [3]

Question 2.

- (a) Derive an expression for $\text{grad} \phi$ in orthogonal curvilinear coordinates (u_1, u_2, u_3) . Hence show that $|\text{grad} u_p| = h_p^{-1}$, $p = 1, 2, 3$ and find an expression for $\text{grad} \phi$ if the transformation is $x = az, y = bx, z = cy$. [6]
- (b) A particle moves along the curve $\vec{r} = (t^3 - 4t)\hat{i} + (t^2 + 4t)\hat{j}$.

- (i) Find the length of the curve it traversed after a time $t = 2$.
- (ii) Find the component of the velocity and acceleration in the direction $\hat{i} + \hat{j}$ after a time $t = 1$. [6]

Question 3.

- (a) State and prove Green's theorem on a simply connected plane region R . [6]
- (b) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = -\frac{y}{x^2+y^2}\hat{i} + \frac{x}{x^2+y^2}\hat{j}$ where the region R is bounded by a circle C of radius a centred at $(0,0)$. [6]

Question 4.

- (a) Evaluate the line integral $\oint_C (yz \, dx + zx \, dy + xy \, dz)$ where C is the curve $x^2 + y^2 = 1, z = y^2$. [6]
- (b) Evaluate $\iiint_V (x^2 + y^2 + z^2) \, dx \, dy \, dz$ where V is the volume of the sphere $x^2 + y^2 + z^2 \leq 1$. [6]

Question 5.

- (a) State Gauss' divergence theorem. Using this theorem, evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where S is the surface of the domain $x^2 + y^2 = a^2$ bounded by the planes $z = 0, z = b$ and $\vec{F} = x\hat{i} - y\hat{j} + z\hat{k}$. [6]
- (b) If $\vec{F} = 2y\hat{i} - z\hat{j} + x\hat{k}$, evaluate $\int_C \vec{F} \times d\vec{r}$ along the curve $x = \cos t, y = \sin t, z = 2 \cos t$ from $t = 0$ to $t = \frac{\pi}{2}$. [6]

Question 6.

- (a) Consider the transformation in the curvilinear coordinate system as $x = \frac{1}{2}(u^2 - v^2), y = uv, z = z$. Determine the scale factors, the unit vectors, the area elements and the volume element. [7]
- (b) Suppose that T^i is a contravariant vector in $\mathbb{R}^2$ and that $(T^i) = (x^2, x^1)$ in the (x^i) system. Calculate (arT^i) in the (arx^i) system under the change of coordinate $arx^1 = (x^2)^2, arx^2 = x^1x^2$. [5]

Question 7.

- (a) Define covariant and contravariant tensors of order two. Show that any covariant tensor of order two can be expressed uniquely as the sum of a symmetric and a skew-symmetric tensor of order two. [6]

- (b) Consider the transformation $x = r \cos \theta, y = r \sin \theta, z = z$. Express ds^2 in the form $ds^2 = \sum_{i,j=1}^3 g_{ij} dx^i dx^j$. Determine the metric $g = \det(g_{ij}), i, j = 1, 2, 3$. Hence find the volume element in the generalized coordinate system. [6]